

See Appendix G for details on the discounting assumptions within the valuation functions.

Should I discount the health incidence as well as the valuation?

You should not discount the health incidence for any of the scenarios mentioned above. Changes in the lag assumptions do not change the total number of estimated deaths, for example, but rather the timing of those deaths. If you discounted the health incidence along with valuation, you would essentially be discounting twice.

Which health endpoints do not occur in the same year as exposure?

In many cases, the health effect from exposure to air pollution occurs shortly after exposure, but there can be a significant lag between exposure and the health effect. The cessation lag can be a matter of hours or days, but some health effects may lag exposure by much longer. If exposure and the health effect do not occur within the same year, it is necessary to discount those benefits back to the analysis year. The only PM_{2.5} health function currently in BenMAP-CE that falls into this category is PM_{2.5}-related premature mortality. Discounting PM-related premature mortality is controversial because the lag structure is unknown, but scientific literature on similar adverse health effects and new intervention studies suggest that premature mortality probably would not occur in the same year as the exposure. (See: Roosli M, Kunzli N, Braun-Fahrlander C, Egger M. 2005. "Years of life lost attributable to air pollution in Switzerland: dynamic exposure-response model." *International Journal of Epidemiology* 34[5]:1029-35.) VSL valuation functions in BenMAP-CE incorporate cessation lags using both 3% and 7% discount rates.

EPA's Science Advisory Board recommends future research to support the development of defensible lag structures and provided a lag structure that could be assumed until additional research has been completed. See Chapter 5 of the PM Regulatory Impact Analysis for more detail on assumed lag structures for PM_{2.5}-related premature mortality (<http://www.epa.gov/ttn/ecas/regdata/RIAs/finalria.pdf>). Some example lag structures from the PM RIA are shown in Figures 7-1 and 7-2 below. Currently, BenMAP-CE does not have the capability to do this type of discounting, so you must discount outside of BenMAP-CE.

Note: Discounting is not necessary for short-term ozone-related premature mortality because it occurs within the analysis year. However, discounting is necessary for premature mortality associated with long-term ozone exposure.

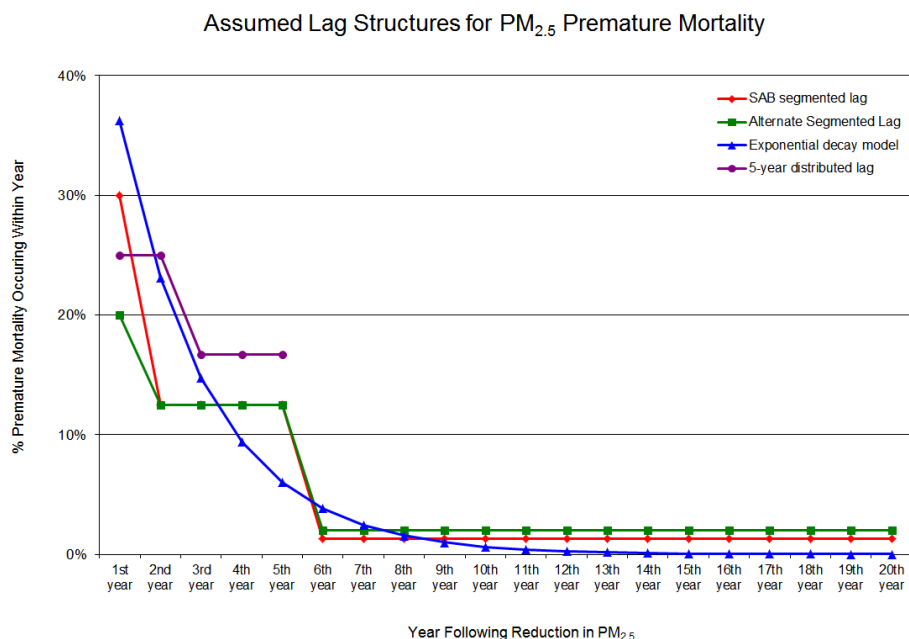


Figure 7-1. Graphical representation of assumed lag structures analyzed in EPA's PM RIA as sensitivity analyses

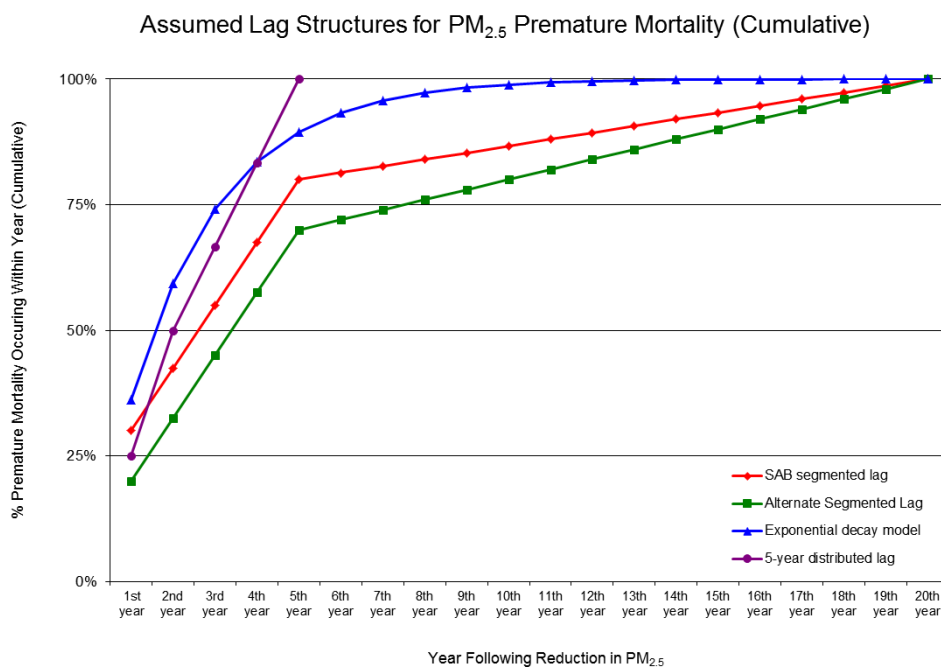


Figure 7-2. Graphical representation of cumulative assumed lag structures analyzed in EPA's PM RIA as sensitivity analyses

7.1.3 Overview of Pooling

For many of the health endpoints (e.g., respiratory hospital admissions), BenMAP-CE contains many different functions from different studies that you could choose to include in your configuration. Combining data from several comparable studies in order to analyze them together is often referred to as meta-analysis. For a number of reasons, it is often impractical or impossible to combine the original data sets. Combining the results of studies provides a second-best way to synthesize information. This is referred to as pooling.

BenMAP-CE allows users to pool the estimated incidence changes predicted by several studies for the same pollutant-health endpoint group combination (e.g., PM_{2.5}-related cardiovascular hospital admissions). It also allows the pooling of the corresponding study-specific estimates of monetary benefits.

Fundamental Concept - Pooling

Pooling refers to the methods of combining of different sets of results generated by multiple health impact functions or valuation functions. For example, BenMAP-CE allows you to pool the estimated incidence changes predicted by several health impact functions for the same pollutant-health endpoint group combination (e.g., PM_{2.5}-related cardiovascular hospital admissions) so that you can present a single value integrating the information generated by all those functions. BenMAP also allows you to pool the economic benefits estimates, e.g. to generate an average valuation from two different studies for ED visits for asthma.

BenMAP-CE includes several pooling methods, which are described in more detail below and in Appendix K.

Why would you want to pool results?

There are two good reasons to pool across study results, one practical and one methodological. Pooling allows you to

- Combine or aggregate multiple study estimates into a single estimate. This combined estimate is easier to report.
- Certain types of pooling—including random effects techniques—account for heterogeneity in the risk estimates reported in the epidemiological literature used to construct the health impact functions you used to calculate incidence.

However, as we discuss below, pooling may not be such a good option if

- You don't know a great deal about the studies used to quantify health impacts; you'll need to know a lot about epidemiological studies used to construct the health impact functions in order to pool properly.

Decision Point - Pooling

When might you want to pool your health impact estimates?

- You have multiple studies of the same health endpoint in the same population group and wish to report a single estimate reflecting the contributions of all the studies.
- You have multiple studies of different (non-overlapping) health endpoints in the same population group that you want to combine to simplify reporting of results (useful for HAs or ED visits).
- You have multiple studies of partially overlapping health endpoints and want to isolate the estimate of a specific health endpoint that appears in one HIF but not the other (via subtraction).

- You think it is important to convey the variability across incidence estimates for a given health endpoint.

BenMAP-CE allows you to pool in five different ways:

- Addition
- Subtraction
- User-assigned weights
- Random Effects
- Fixed Effect.

The examples that follow demonstrate each of these methods (Fixed and Random Effects are combined in one example.) In each of the examples below, the distribution of estimated health impacts are represented by a normally distributed probability density function (PDF) shaped as a bell curve. In each PDF, the mean health impact estimate is represented by a dashed line. While we illustrate these examples using a normally distributed PDF for ease of presentation, BenMAP-CE can accommodate several other types of distributions (e.g. Weibull, Triangular).

BenMAP-CE Use Example

Example scenarios for pooling health impact estimates:

- You ran two different HIFs for HA, All Cardiovascular for ages 65-99. They report different risk estimates and standard errors, and you'd like to combine them into a single estimate.
- You ran an HA, Ischemic Heart Disease HIF for ages 65-99 and an HA, Dysrhythmia HIF for ages 65-99 and want to combine the results into a single value for HA, cardiovascular outcomes.
- You ran an HA, All Cardiovascular HIF for ages 65-99 and an HA, All Cardiovascular, *less stroke* HIF for ages 65-99 and want to use the two functions to estimate the number of stroke-related HAS.